



Sri Lanka Institute of Information Technology

Network Technology Project

Topic Assessment Form

PROJECT TITLE	Industrial IoT -Based Compressed Air Leakage Detection and Energy Assessment
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PROJECT NUMBER		(will be assigned by the lecturer-in-charge)
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PROJECT GROUP MEMBER DETAILS: (Please start with group leader's details)

	STUDENT NAME	STUDENT NO.	CONTACT NO.	EMAIL ADDRESS (SLIIT mail address)	SIGNATURE
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SUPERVISOR	
	(will be assigned by the lecturer in charge)

Name		
ACCEPTANCE BY Lecturer		
Name	Signature	Date

PROJECT DETAILS

Background

Compressed air is one of the most expensive energy sources used in manufacturing industries. It powers pneumatic cylinders, robotic systems, packaging machines, conveyor systems, CNC machines, and production equipment. Studies consistently report that air leakage can waste a significant fraction of compressed air production, increasing electricity consumption, maintenance costs, and carbon emissions. Traditional inspection methods often rely on **manual ultrasonic testing**, which is time-consuming and may require production interruptions. Recent research has explored IoT sensing and **machine learning for leak detection**, but many approaches primarily identify the existence of a leak rather than helping maintenance teams prioritize repairs and estimate operational impact.

Proposed Solution

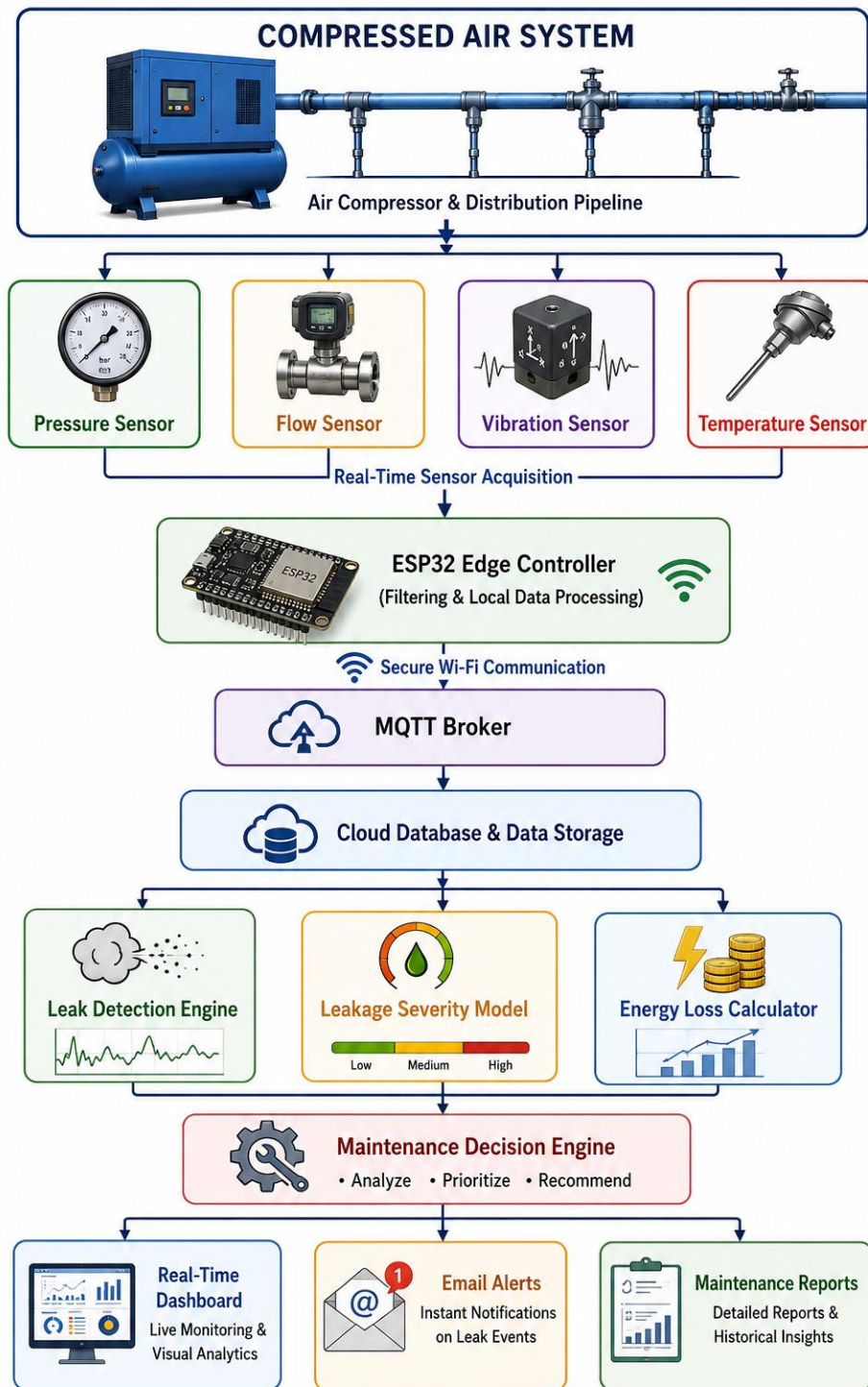
The proposed system develops a low-cost **Industrial IoT framework** that continuously monitors compressed air pipelines using pressure, airflow, **vibration**, and compressor operating data. Sensor data are collected by ESP32 edge devices and transmitted securely through MQTT to a cloud platform.

Instead of simply detecting leaks, the system performs intelligent leakage assessment by:

- Detecting abnormal operating conditions.
- Estimating leakage severity.
- Estimating compressed air loss.
- Estimating electrical energy loss.
- Estimating financial loss.
- Prioritizing maintenance based on risk.

A web dashboard provides real-time monitoring, maintenance recommendations, historical analysis, and alert notifications.

The proposed solution is intended for small and medium-sized manufacturing industries that require an affordable predictive maintenance system without expensive industrial monitoring equipment.



Main Objective

To design and develop an Intelligent Industrial Internet of Things framework capable of continuously monitoring compressed air systems, detecting leakage conditions, assessing leakage severity, estimating energy and financial losses, and supporting predictive maintenance decisions through real-time monitoring and intelligent analytics.

Sub Objectives

SO1

Design and implement a multi-sensor Industrial IoT data acquisition system capable of continuously monitoring pressure, airflow, vibration, and temperature parameters within a compressed air distribution network.

SO2

Develop a secure Industrial IoT communication framework using ESP32, Wi-Fi, MQTT, and cloud technologies for reliable real-time transmission, storage, and management of sensor data.

SO3

Develop an intelligent leakage assessment algorithm capable of identifying abnormal compressor behavior, estimating leakage severity, calculating compressed air loss, estimating energy consumption, and predicting the financial impact of leakage.

SO4

Develop a web-based monitoring and decision-support platform that provides real-time visualization of compressor performance, maintenance recommendations, historical analysis, and automated maintenance alert

WORKLOAD ALLOCATION (Please provide a brief description about the workload allocation)

MEMBER 1	Praveena Dasanayake- Project Management, Industrial System Analysis & System Architecture Design
Responsibilities 1. Project Management • Prepare project timeline, milestones, and task schedule. • Monitor project progress and coordinate team activities. • Identify project risks and manage requirements. 2. Industrial System Analysis • Study industrial compressed air systems and leakage problems. • Analyze existing leakage detection methods and limitations. • Identify system requirements and required monitoring parameters. 3. System Architecture Design • Design the overall Industrial IoT system architecture. • Define communication between hardware, cloud, analytics, and dashboard modules. • Prepare system workflow and data flow diagrams. 4. System Integration Planning • Define integration and testing procedures. • Coordinate final system testing and evaluate overall functionality. Deliverables • Project Plan and Timeline • Requirement Analysis Report • System Architecture Design • Data Flow Diagram • Integration Testing Pla	

MEMBER 2	Gamage Sithumini - Cloud Infrastructure & Industrial IoT Communication
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Responsibilities**1. IoT Communication Development**

- Design communication between ESP32 devices and cloud platform.
- Implement Wi-Fi and MQTT-based data transmission.
- Configure MQTT broker for real-time communication.

2. Cloud Infrastructure Development

- Design cloud data storage structure.
- Configure database for sensor data management.
- Integrate cloud services with other system modules.

3. IoT Security Implementation

- Implement secure communication using encryption methods.
- Configure device authentication and data protection mechanisms.
- Evaluate communication security.

4. Network Performance Evaluation

- Test communication reliability, latency, and data transmission efficiency.
- Troubleshoot connectivity issues.

Deliverables

- IoT Communication Architecture
- MQTT Communication Module
- Cloud Database Setup
- Security Implementation
- Network Performance Report

MEMBER 3	Anuradha Jayashanka - Embedded Hardware Development & Sensor Integration
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Responsibilities

1. Hardware Design

- Select suitable sensors for pressure, airflow, temperature, and vibration monitoring.
- Design hardware connections with ESP32.
- Prepare circuit design.

2. Embedded System Development

- Develop ESP32 firmware for sensor data collection.
- Integrate sensors with the microcontroller.
- Implement basic edge-level data processing.

3. Prototype Development

- Build the compressed air monitoring prototype.
- Create controlled leakage conditions for testing.
- Perform hardware troubleshooting.

4. Sensor Testing

- Calibrate sensors and validate measurements.
- Evaluate sensor accuracy and reliability.

Deliverables

- ESP32 Prototype
- Sensor Integration System
- Firmware
- Circuit Diagram
- Sensor Testing Report

MEMBER 4	Adhithya Mapa - Leakage Analytics & Decision Support System
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Responsibilities

1. Leakage Detection Analysis

- Analyze sensor data to identify leakage conditions.
- Develop leakage detection methods.
- Classify leakage severity levels.

2. Energy and Cost Analysis

- Estimate compressed air loss caused by leakage.
- Calculate energy wastage and financial impact.
- Generate maintenance recommendations.

3. Dashboard Development

- Develop web-based monitoring dashboard.
- Display real-time sensor data and leakage status.
- Visualize historical performance data.

4. Alert and Reporting System

- Implement leakage alerts and notifications.
- Generate system reports for maintenance decisions.
- Test application performance.

Deliverables

- Leakage Detection Module
- Energy Loss Analysis
- Monitoring Dashboard
- Alert System and Application Testing Report

